

medfilt2 Function Test Results

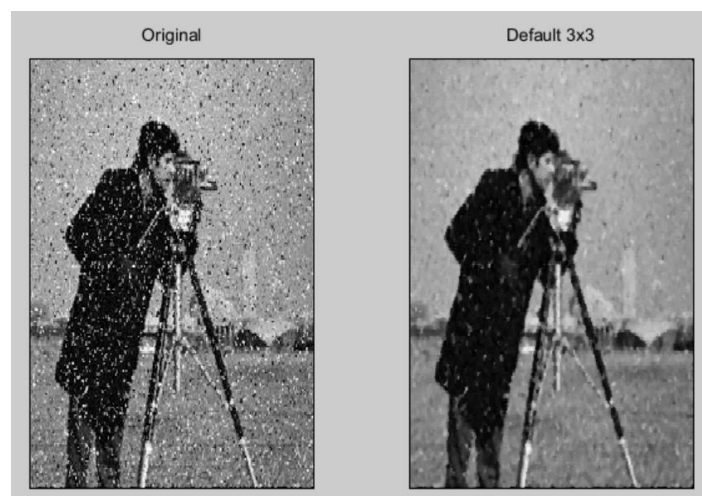
Test Case 1: Default Parameters

Input Image: Cameraman image

Parameters:

- Window Size: 3×3
- Padding: Zero (default)

Output:



Observation: Default median filtering reduces small noise while preserving major edges.

Test Case 2: Square Window [5×5]

Parameters:

- Window Size: 5×5
- Padding: Replicate

Output:



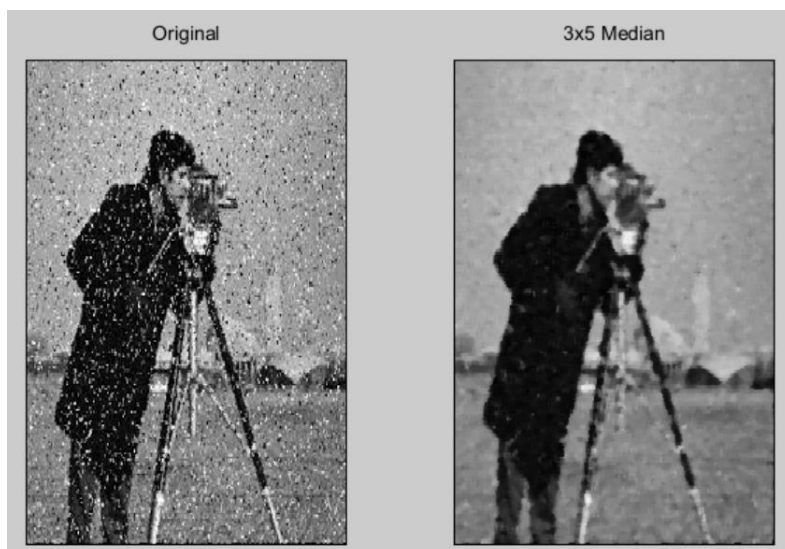
Observation: Larger window provides stronger smoothing but removes more image detail.

Test Case 3: Rectangular Window [3×5]

Parameters:

- Window Size: 3×5
- Padding: Replicate

Output:



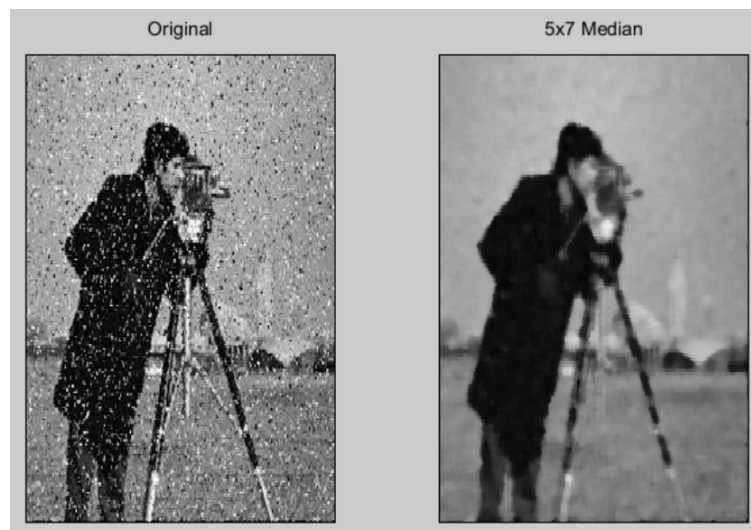
Observation: Noise reduction is stronger along the horizontal direction due to the rectangular neighborhood.

Test Case 4: Rectangular Window [5×7]

Parameters:

- Window Size: 5×7
- Padding: Replicate

Output:



Observation: Produces significant smoothing and suppresses fine image structures.

Test Case 5: Zero Padding

Input Matrix:

10 20 30

40 50 60

70 80 90

Parameters:

- Window Size: 3×3
- Padding: Zero

Output:

0. 20. 0.

20. 50. 30.

0. 50. 0.

Observation: Boundary pixels are influenced by surrounding zeros.

Test Case 6: Replicate Padding

Parameters:

- Window Size: 3×3
- Padding: Replicate

Output:

20. 30. 30.

40. 50. 60.

70. 70. 80.

Observation: Border values are preserved by extending edge pixels.

Test Case 7: Circular Padding

Parameters:

- Window Size: 3×3
- Padding: Circular

Output

50. 50. 50.

50. 50. 50.

50. 50. 50.

Observation: Opposite image boundaries contribute to neighborhood values.

Test Case 8: Symmetric Padding

Parameters:

- Window Size: 3×3
- Padding: Symmetric

Output:

20. 30. 30.

40. 50. 60.

70. 70. 80.

Observation: Image borders are mirrored including edge pixels.

Test Case 9: Reflect Padding

Parameters:

- Window Size: 3×3
- Padding: Reflect

Output:

40. 40. 50.

50. 50. 50.

50. 60. 60.

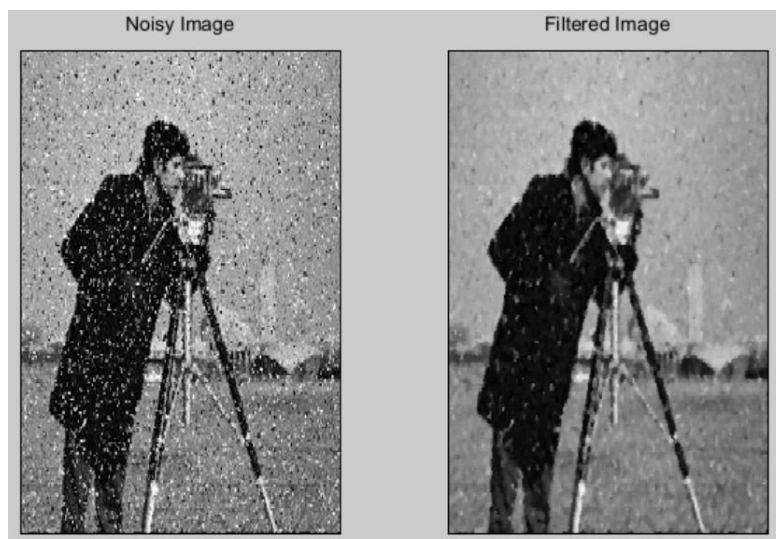
Observation: Image borders are mirrored excluding edge pixels.

Test Case 10: Salt-and-Pepper Noise Removal

Parameters:

- Window Size: 3×3
- Padding: Replicate

Output:



Observation: Impulse noise is effectively removed while maintaining object boundaries.

Test Case 11: Extreme Noise Pixel

Input Matrix:

10 20 30
40 255 50
60 70 80

Output:

20 30 30
40 50 50
60 70 80

Observation: The outlier value (255) is replaced by the neighborhood median.

Test Case 12: Filter Larger Than Image

Input Matrix:

10 20
30 40

Parameters:

- Window Size: 5×5
- Padding: Replicate

Output:

20. 20.
30. 30.

Observation: Padding allows filtering even when the window exceeds image dimensions.

Test Case 13: Single Pixel Image

Input:

150

Output:

150

Observation: The image remains unchanged because all neighborhood values are identical.

Test Case 14: Invalid Padding Option

Input Parameters

Padding = "abc"

Output: "Invalid padding option"

Observation: Function correctly rejects unsupported padding methods.

Test Case 15: Even Window Size

Input Parameters

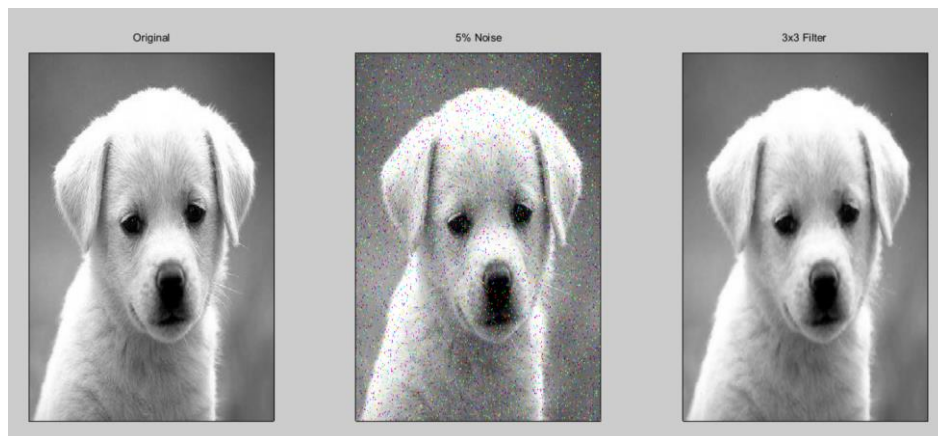
Window Size = [4 4]

Output: "Window dimensions must be odd"

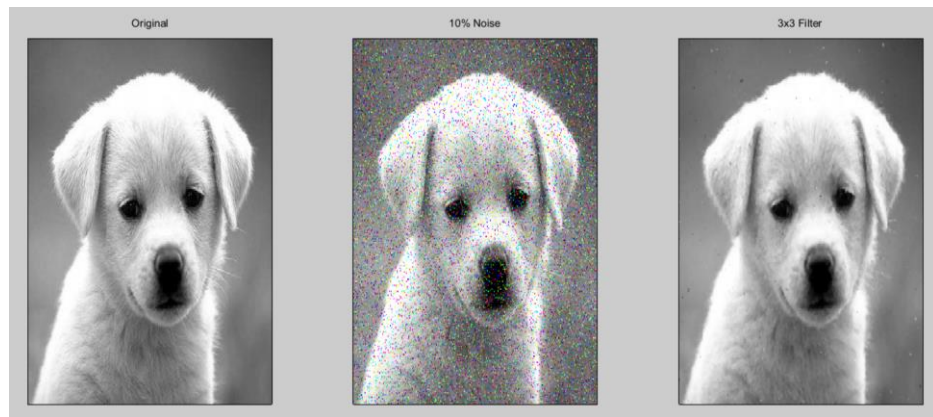
Observation: Median filtering requires odd-sized neighborhoods and the validation works correctly.

Additional Test Cases:

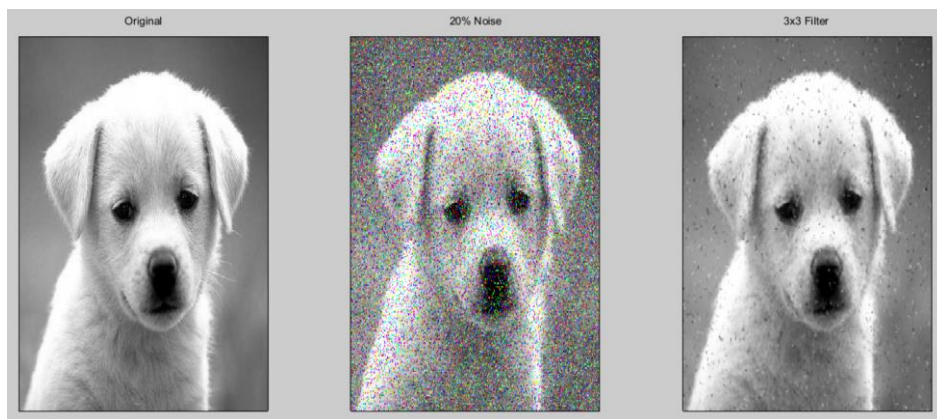
1. Low Salt-and-Pepper Noise (5%)



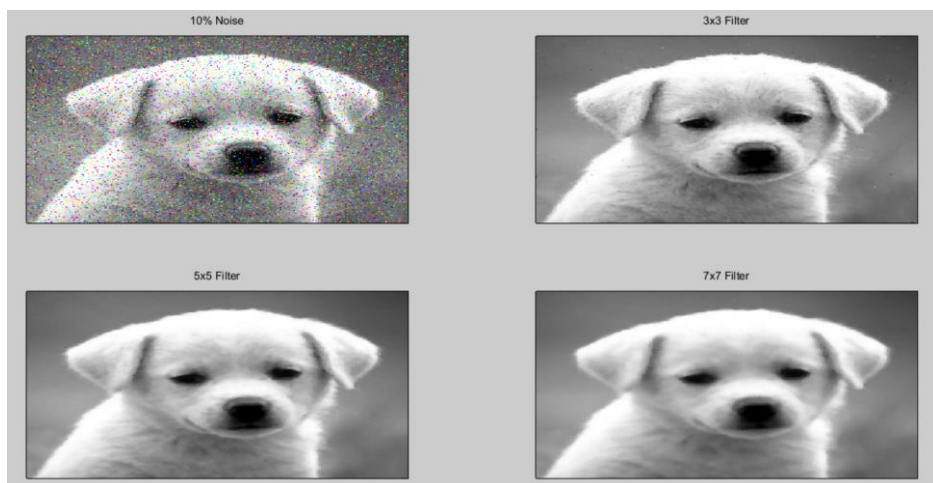
2. Medium Salt-and-Pepper Noise (10%)



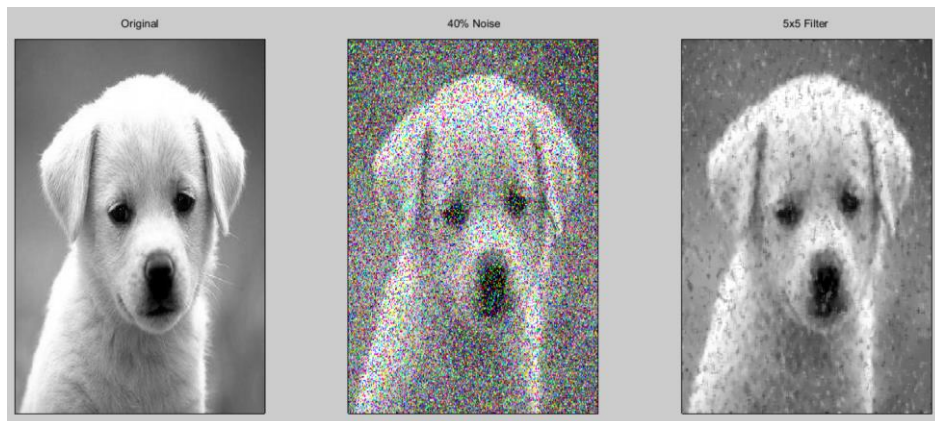
3. High Salt-and-Pepper Noise (20%)



4. Noise Level Fixed, Filter Size Comparison



5. Extreme Salt-and-Pepper Noise (40%)



6. Edge Preservation

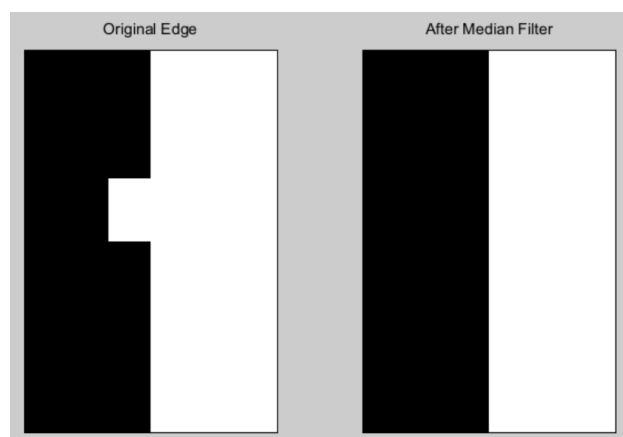
Input Matrix

```
0 0 0 255 255 255
0 0 0 255 255 255
0 0 255 255 255 255
0 0 0 255 255 255
0 0 0 255 255 255
0 0 0 255 255 255
```

Parameters:

- Window Size: 3×3
- Padding Method: Replicate

Output:



Observation: Median filtering successfully removes isolated outlier pixels while preserving major intensity boundaries.